

The zero-sum sheath stabiliser — what it actually does

ported from SOLPS-ITER b2stbc_stab_coeff_sheath_te/ti/ni

1. The problem. With $\Phi = 3T_e$ at the wall, the boundary n_e and T_e overshoot to *negative* in a single time step, in 2–3 cells. The step is too big for that row.

2. Key fact about implicit solving. An implicit step has two separate ingredients:

residual b “how wrong am I?” $\rightarrow$ decides **WHERE** the answer is ($b = 0$)
Jacobian A “how do I move?” $\rightarrow$ decides **HOW FAST** I get there

The converged state is where $b = 0$. **A does not appear in that condition.** So we may stiffen A as much as we like — it cannot move the answer.

3. The trick. Add to the boundary source a term that is *zero when nothing is changing*:

$$S^{\text{stab}} = \alpha \Pi (T^{\text{old}} - T^{\text{new}}), \quad \Pi = \text{the sheath flux factor already in the term}$$

At steady state $T^{\text{new}} = T^{\text{old}}$, so $S^{\text{stab}} \equiv 0$: it contributes *nothing* to b . But its derivative $\partial S^{\text{stab}} / \partial T^{\text{new}} = -\alpha \Pi$ is *not* zero, so it adds to A .

4. Worked example (all of it, in three lines). Take a scalar boundary equation with a source Q and a sink rate k , solved implicitly:

$$\frac{T^{\text{new}} - T^{\text{old}}}{\Delta t} = Q - k T^{\text{new}} \quad \Rightarrow \quad \Delta T = \frac{\overbrace{Q\Delta t - k\Delta t T^{\text{old}}}^{\text{numerator}}}{1 + k\Delta t}$$

Now add $S^{\text{stab}} = \alpha k (T^{\text{old}} - T^{\text{new}})$:

$$\Delta T = \frac{Q\Delta t - k\Delta t T^{\text{old}}}{1 + k\Delta t + \alpha k\Delta t}$$

The numerator is identical. Only the denominator grew.

- Steady state: set $\Delta T = 0 \Rightarrow T^* = Q/k$ — **the same with or without α .**
- Step size: reduced by $\frac{1 + k\Delta t}{1 + k\Delta t + \alpha k\Delta t}$ — **the overshoot is damped.**

5. In the code it is one line. Since the physical Jacobian entry carries $(\gamma_{\text{sheath}} - 1)$, adding $\alpha \Pi$ to it is the same as writing

$$(\gamma_{\text{sheath}} - 1 + \alpha) \text{ in the Jacobian,} \quad (\gamma_{\text{sheath}} - 1) \text{ in the residual.}$$

Applied to the n_e , T_i , T_e boundary rows; nothing off-diagonal, no other equation.

Careful: scaling *both* residual and Jacobian would simply change γ_{sheath} — a different physics problem, a different answer. The whole point is that **only the Jacobian** is touched.

6. Analogy. It is a **damper**, not a spring. In a mass–spring system the equilibrium is set by the spring; adding a dashpot changes only how it settles, never where. Here α is the dashpot on the boundary temperature.

7. Why α is not a fudge factor. α is dimensionless — a multiple of the sheath transmission coefficient already present. SOLPS ships it, recommends $\alpha = 1$ –100, and defaults to 0; $\alpha = 0$ reproduces our previous results bit-for-bit. It damps the **transport** rows, not the momentum condition (SOLPS ships none for that) — matching our failure, where n_e and T_e go negative *together* in the same cells.

8. Result. Drift-enabled run: previously died at ≤ 649 steps $\rightarrow$ **now past 1100 and converging** (min. density flat, $|v_{E,n}|$ falling, successive drops in min. wall T_e shrinking).

Caveat: SOLPS iterates inside a time step, so there the term vanishes before the step is accepted and is purely a convergence aid. JOREK takes one solve per step, so T^{old} is the previous *time step* and genuine transients are slowed too. The stationary state is unaffected either way.